

Exercise 9.1

Determine whether each of the following systems are controllable:

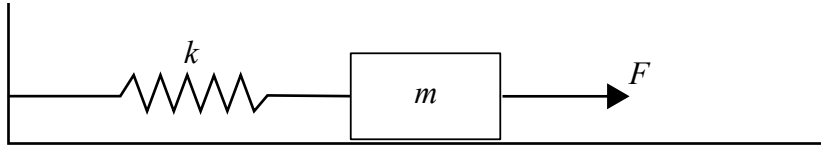
$$\begin{aligned} \dot{x} &= \begin{bmatrix} -1 & 2 \\ 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u, & \dot{x} &= \begin{bmatrix} -1 & 0 \\ 2 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u, \\ \dot{x} &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, & \dot{x} &= \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u. \end{aligned}$$

Exercise 9.2

Consider the mass-spring-damper system illustrated below with dynamics

$$\begin{bmatrix} \dot{p} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} p \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F,$$

where $m = 1$, $k = 2$, and $c = 3$.



1. Transform the system to discrete time, with a sampling time of 0.05 s. (Use the function `c2d`)
2. Calculate a control sequence $u(0), u(1)$ for the discrete time system such that $x(2) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, when $x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
3. Verify the control by simulating the system in MATLAB. (Hint: Use a for-loop)

Exercise 9.3

Consider the following system

$$\dot{x} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u.$$

1. Find a state space transformation that brings the system into controllable canonical form.
2. Based on the system description in controllable canonical form, find a state feedback $u = Fx$, such that the characteristic polynomial of the closed loop system is $s^3 + 3s^2 + 6s + 4$. Note that the feedback must be given in the original states.
3. Based on the system description in controllable canonical form, find a state feedback $u = Fx$, such that the poles of the closed loop system are -1 , -2 , and -3 .
4. Verify both controller designs by computing the eigenvalues of $A + BF$.